EduSense — Architecture & Methodology Specification

Problem 13: Dictionary and Thesaurus-Based Word Sense Disambiguation for Educational Transcripts

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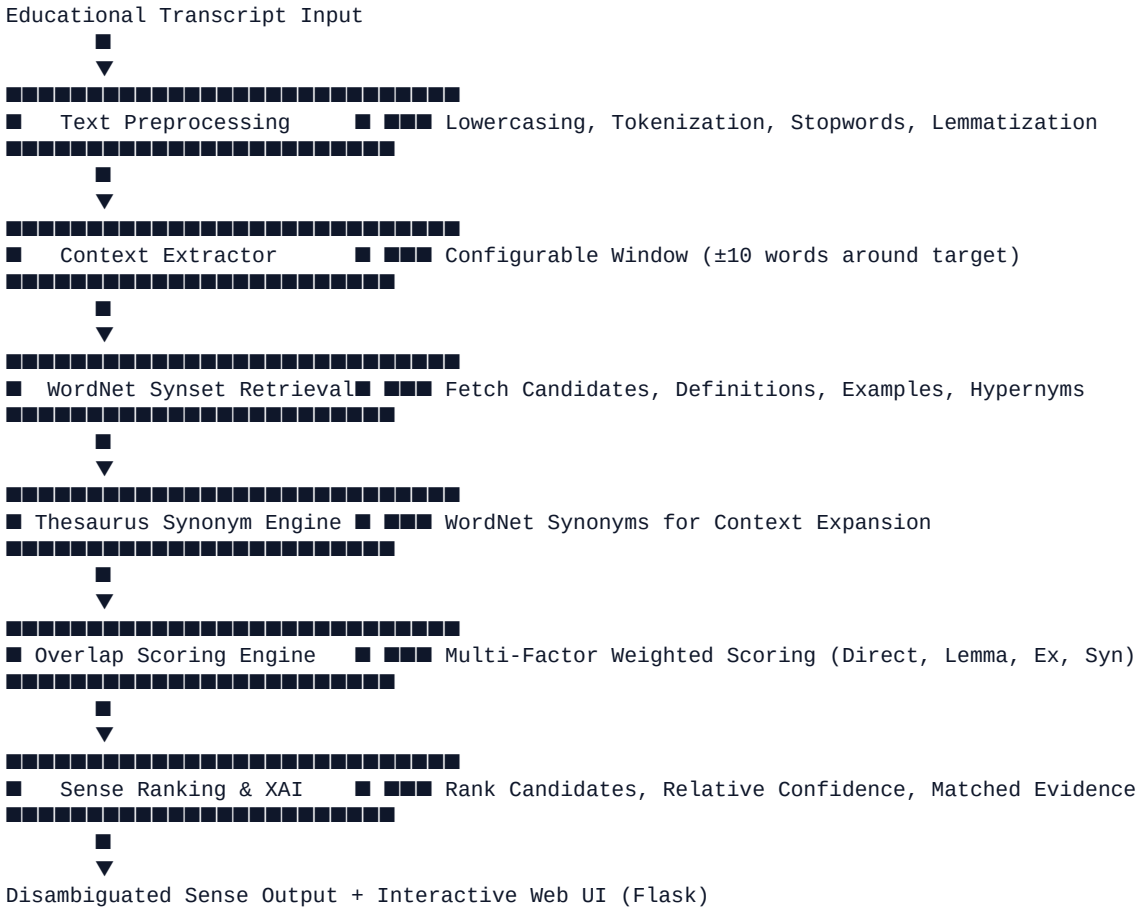
1. Problem Definition & Academic Context

Educational lecture audio transcripts contain polysemous academic vocabulary whose meaning varies critically across academic domains (e.g., Geology vs Economics vs Biology). For instance, the target word *'bank'* can signify sloping land beside water in earth science or a financial deposit institution in macroeconomics. Determining the correct intended meaning requires querying structured lexical databases such as WordNet and scoring contextual vocabulary overlaps against definition glosses and example sentences without relying on proprietary, black-box Large Language Model APIs.

2. System Architecture & Processing Pipeline



3. Mathematical Scoring Formulation

EduSense computes a transparent, multi-factor overlap score for every candidate WordNet synset *s*:

$$\text{Score}(s) = \alpha \cdot \text{DirectOverlap}(C, G_s) + \beta \cdot \text{LemmaOverlap}(C_{lemma}, G_{s,lemma}) + \gamma \cdot \text{ExampleOverlap}(C_{lemma}, E_{s,lemma}) + \delta \cdot \text{ThesaurusOverlap}(S_{ctx}, G_{s,lemma}) + \varepsilon \cdot \text{HypernymOverlap}(C_{lemma}, H_s)$$

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Where:
• C: Processed context tokens (excluding target word)
• G_s: Gloss definition tokens of candidate synset s
• E_s: Example sentence lemmas of synset s
• S_ctx: WordNet synonym expansion of context words
• H_s: Hypernym and hyponym hierarchy tokens of synset s
Default Weights: α = 1.5, β = 1.2, γ = 1.0, δ = 0.7, ε = 0.5
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4. Relative Confidence Metric

Relative confidence represents how clearly the predicted synset dominates rival candidate senses without claiming uncalibrated probabilistic certainty:

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Relative Confidence(s*) = Top Score(s*) / ∑_{s ∈ Synsets} Score(s)
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5. Modular Subsystem Deliverables

Module Path	Subsystem Responsibility	Key NLP Technique
src/preprocessing.py	Text normalization & lemmatization	Penn Treebank POS, WordNetLemmatizer
src/context_extractor.py	Window extraction & target markup	Configurable windowing (±5 to ±25)
src/wordnet_engine.py	WordNet querying & thesaurus expansion	Synsets, glosses, examples, hypernyms
src/models.py	Dictionary & Thesaurus WSD model	Multi-factor scoring, POS tagger model
src/pipeline.py	Execution pipeline & benchmark runner	End-to-end WSD execution & testing
app.py	Interactive web application & REST API	Flask, Bootstrap 5, Chart.js XAI UI